Curriculum Vitae
ABDELRAHMAN A. ELABD

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Education

2022 - Present — Ph.D., Physics
University of Washington
Seattle, WA 98195

2017 - 2021 — B.A., Physics + Economics
University of Pennsylvania - Mayor's Scholar 2017-2021
Philadelphia, PA 19104

Research Interests

- **Hardware-accelerated ML for real-time inference**
- **Generative models for fast simulation**
- **Autonomous control**
- **Quantum information**

Publications

1. A. Elabd, V. Razavimaleki, S. Y. Huang, J. Duarte, M. Atkinson, S. C. Hsu et al. "Graph Neural Networks for Charged Particle Tracking on FPGAs," [InspireHEP, 39 citations].

Research (Current)

Aug 2025 - Present — **Search for Quantum Entanglement and Bell Nonlocal States in $Z \longrightarrow \tau\tau$**
- Use ML and Monte Carlo to quantify Bell nonlocality in $Z \longrightarrow \tau\tau$ decays at the LHC.
- Train foundation model to perform neutrino/hidden-particle reconstruction.

Mar 2025 - Present — **WATCHEP Fellow**
- Western Advanced Training in Computational High-Energy Physics (WATCHEP).
- Use generative models to emulate particle-detectors during high-energy collision events.
- Developing software package for full physics-simulation and detector-emulation pipeline.
- Deploy multi-node training of diffusion and flow-matching models on Perlmutter supercomputer at LBNL/NERSC.

Feb 2025 - Present — **ATLAS Pixel Detector DAQ development**
- Software-hardware co-design of heterogeneous algorithms for ATLAS Pixel Detector data-acquisition (DAQ) and trigger system.
- Develop, simulate, deploy co-processor FPGA designs with embedded Linux kernels.

Sep 2024 - Present — **Real-time inference on FPGAs for thin-film deposition with RHEED**
- Developed FPGA design of sequential, multiple-model ML inference chain for real-time characterization of materials during growth via thin-film deposition.
- Enables research on high-speed molecular and plasma dynamics of thin-film deposition.
- Basis for future development in real-time optimal control of thin-film deposition.
- Supervise ML workflow including data augmentation and simulation, pruning, quantization-aware training, semi-supervised learning, and neural architecture search.

Awards and Funding

2025-2027	WATCHEP Fellowship
2021	IRIS-HEP Fellowship
2017-2021	Mayor's Scholarship

Contributed Talks

Sep 2025	*FastML for Real-Time RHEED Inference* Fast Machine Learning for Science 2025, Zurich, Switzerland
May 2022	*Graph Neural Networks for Charged Particle Tracking on FPGAs* Connecting the Dots 2022, Princeton, NJ

Blogs and Outreach

- **Hackathon Organizer:** U. Washington chapter of HDR FAIR ML Challenge 2025/2026
- **Blog Author:** A3D3 at FastML 2025
- **Instructor:** A3D3 AI Summer Camp 2025
- **Instructor:** Quarknet CMS Masterclass, U. Washington: 2022 - 2025

Research (Past)

Jun - Sep 2024 **Reinforcement learning for accelerated calibration of trapped-ion qubits**

- Trained reinforcement learning (RL) agent to calibrate two-qubit entangling gates using sparse, stochastic, (quantum-)measurement-based adaptive feedback.
- Developed noise-model simulation of Ca-40^+ optical trapped-ion qubits.

Mar 2024 - Jun 2024 **AI- and hardware-accelerated readout of superconducting qubits**

- Implemented, tested FPGA ML algorithms for scalable, low-latency readout of superconducting qubits with the open-source QICK RFSoC qubit controller.

Apr 2024 **Founder: UW Quantum Computing Hardware Journal Club**

Jan 2021 - Apr 2022 **IRIS-HEP Fellow**: Graph Neural Networks for Charged Particle Tracking on FPGAs

- **First-author** in *Frontiers in Big Data*, 2022.
- Developed hls4ml.pyg_to_hls toolkit for automatic translation of Graph Neural Networks (GNNs) into optimized High-Level Synthesis (HLS) FPGA designs.

Industry Experience

Aug 2021 - Oct 2023 **Jr. Algorithm Engineer**
TDI Technologies, Inc.
Philadelphia Naval Yard

- Developed, deployed ML algorithms for systems monitoring aboard US Navy vessels.
- Developed, maintained pipeline for ingestion of signals data from shipboard sensors.
- Provided documentation and testing instructions for enlisted personnel and engineers.
- Security clearance: Secret.

Computation

- **Full-stack FPGA:** (System)Verilog, VHDL, HLS, hls4ml, Vivado, ModelSim
- **Software:** Python, C++, Julia, Mathematica, Linux, Git
- **Reinforcement Learning:** OpenAI Gym
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn
- **High-Performance Computing:** SLURM, Horovod, torch.DDP, Lightning, SSH/PuTTy, Docker

- **Numerical simulation**
- **Quantum Computing:** QICK, Azure Quantum, qiskit, QuTiP

Conferences

Sep 2025	Fast Machine Learning for Science 2025 ETH Zurich, Switzerland
Aug 2025	A3D3 All-Hands Meeting California Institute of Technology, Pasadena, CA
June 2025	WATCHEP Summer School University of California, Berkeley
April 2025	ATLAS Pixel Week CERN, Geneva, Switzerland
Sep 2024	NSF Harnessing the Data Revolution (HDR) Ecosystem Conference University of Illinois at Urbana-Champaign, Champaign, IL
Jul 2024	US Quantum Information Science Summer School Quantum Science Center, Oak Ridge National Laboratory
May 2022	Connecting the Dots 2022 Princeton University, Princeton, NJ

Teaching (TA)

Winter 2025	PHYS522 Implementations in Quantum Computing
Autum 2024	PHYS521 Quantum Information Theory
Autum 2024	PHYS433 Nuclear and Particle Laboratory
Autumn 2023	PHYS434 Computational Data Analysis
Spring 2023	PHYS417 Neural Network Methods for Signals
(Multiple)	PHYS334 Advanced Laboratory Electronics
(Multiple)	PHYS232 Computational Physics

Notable Coursework

Winter 2025	EE271	Intro to FPGAs
Winter 2024	PHYS576B	Implementations in Quantum Computing
Autumn 2023	PHYS521	Quantum Information Theory
Spring 2023	PHYS564	General Relativity
Spring 2021	ECON224	Statistical Learning and Causal Inference
Spring 2021	ECON212	Game Theory
Spring 2020	PHYS585	Theoretical and Computational Neuroscience
Autumn 2019	STAT430	Probability
Spring 2019	PHYS364	Electronics and Digital Design
Spring 2018	LING105	Intro to Formal Linguistics